GADILA PRAVEEN KUMAR

Senior SDET | Test Automation Architect | AI-Driven Quality Engineering

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PROFESSIONAL SUMMARY

Senior SDET and test automation architect with 11+ years designing hybrid frameworks (Playwright, Selenium, REST Assured, Spring Boot) and AI-driven quality engineering solutions across Web, API, Database, ETL and Mobile testing in Investment Banking, Commodity Insights and Insurance. Built a RAG agent on Azure OpenAI that generates test cases from requirements documents, cutting test-design time by 80%, and an AI visual validation agent that cut manual UI validation effort by 95%. Developed a generic browser-use UI automation agent and a Playwright MCP + GitHub Copilot workflow that generates scripts from skill files. Led 4 automation engineers and drove CI/CD integration, code reviews and framework standards.

CORE SKILLS

Test Automation Framework Design | Playwright | Selenium WebDriver | Java | TestNG | Cucumber BDD | REST Assured | API Testing | Cross-Browser Testing | Page Object Model (POM) | CI/CD | Jenkins | Docker | Kubernetes | AWS | BrowserStack | Agile/Scrum | SDLC | STLC | Regression Testing | Functional Testing | Smoke & Sanity Testing | Performance Testing | SQL | Git | Maven | JIRA | Test Strategy & Planning | Defect Management | Quality Engineering | Team Mentoring | Code Review | Data-Driven Testing | Hybrid Framework | AI-Augmented Testing | Model Context Protocol (MCP) | GitHub Copilot

AI / Agentic Engineering: Azure OpenAI, Azure Embeddings, RAG (Retrieval-Augmented Generation), Agentic AI, LLM Agents, LLM Integration, Prompt Engineering, Model Context Protocol (MCP), browser-use, GitHub Copilot

Test Automation: Test Automation Framework Design, Hybrid & Data-Driven Frameworks, Page Object Model (POM), Playwright (incl. Playwright MCP), Selenium WebDriver, TestNG, Cucumber BDD, pytest-BDD, REST Assured, Spring Boot, Cross-Browser Testing

Languages: Java, Python, SQL

Data & Cloud: Databricks, PySpark, Snowflake, Oracle, MySQL, SQL Server, GCP, Microsoft Azure, AWS, Docker, Kubernetes

CI/CD & Tools: CI/CD, Jenkins, Git, Maven, BrowserStack, JMeter, LoadRunner, JIRA, HP ALM, RTC

Testing Types: API Testing, Regression, Functional, Smoke & Sanity, Performance, ETL / Source-to-Target Validation, Database Testing, Mobile Testing

Domains: Investment Banking, Commodity Insights, Insurance

KEY PROJECTS

AI Test Case Generation Agent (RAG) — S&P Global, Apollo

- Built a RAG pipeline using Azure OpenAI embeddings over requirements documents and user stories, retrieving relevant context to auto-generate test cases, reducing manual test-design effort by 55% for new feature releases.

AI Visual Validation Agent — S&P Global, Apollo

- Designed an AI-powered visual validation agent using Azure OpenAI to analyze page screenshots for rendering, content accuracy, and UI integrity across 200+ scenarios, cutting manual UI validation effort by 95%. Received the Innovation Award for this framework.

Cloud API Compare Tool — S&P Global, Apollo

- Built an automated multi-environment REST API comparison tool with an interactive web UI, cutting cross-environment validation time by 60% for QA and Dev teams; awarded Star of the Quarter.

AI-Driven UI Test Automation Framework (Auto-UI) — S&P Global

- Built **Auto-UI**, an AI-driven test authoring framework where reusable GitHub Copilot skill files guide Copilot, through Playwright MCP, to explore the live application and generate locators, page objects, and BDD step definitions. This replaces manual script writing with a repo generated from natural-language commands.

Generic AI-Powered UI Automation Agent — S&P Global

- Developed a generic AI-powered UI automation agent using browser-use and Azure OpenAI, enabling users to execute complex web tasks (e.g., login, search, filter by conditions) via free-form natural language prompts, with automated validation of test scenarios and webpage states, and generated detailed execution reports with screen-recorded video evidence.

PROFESSIONAL EXPERIENCE

Senior SDET, Test Automation Engineer — S&P Global *Jan 2024 – Present*

Skills: Azure OpenAI, Selenium, Playwright, Playwright MCP, TestNG, Java, Jenkins, BrowserStack, GitHub Copilot

Apollo: Platts market data, news, analytics, and pricing platform for energy, commodities, finance, and trading professionals.

- Architected and developed an enterprise AI-driven test automation framework using Azure OpenAI, LLMs, and Playwright for visual validation and intelligent defect detection, reducing manual UI validation effort by 95%.
- Built proof-of-concept agentic workflows for autonomous test execution and defect reporting, cutting result-triage time by 60%.
- Built and integrated a Playwright MCP (Model Context Protocol) server enabling GitHub Copilot agent mode to autonomously drive, inspect, and generate UI test automation for Platts Connect (React SPA), cutting test authoring time by 50%
- Authored a persistent copilot-instructions.md skills/context file encoding framework conventions, page-object patterns, and coding standards, auto-updated on PR merge via GitHub Actions — enabling GitHub Copilot agent mode to generate new framework code consistent with team standards
- Built API automation suite with REST Assured and Python, reducing API regression time by 60%
- Established Jenkins/GitHub Actions CI/CD pipelines with parallel execution, cutting build feedback from 4 hours to 45 minutes
- Built source-vs-target (API/UI) comparison automation across 15+ modules with detailed per-module summary reporting.
- Built Selenium automation for Web application testing, maintaining full regression coverage (API, UI, DB) each release.
- Set up Jenkins CI/CD pipelines integrated with BrowserStack for parallel execution, cutting pipeline runtime by 65%.
- Managed a team of 4 automation engineers, delivering automation solutions on schedule (Best Team recognition).

ADW (Advanced Data Workspace), from Aug 2026: Databricks, Snowflake, GCP, and Python for data/table validation.

- Built automated data-validation checks in Databricks/PySpark (null checks, duplicate checks, row-count mismatches, source-vs-target comparison) connecting Snowflake and GCP dataframes, cutting manual validation effort by 50%.

Senior Associate, Automation Engineer — State Street India *Jul 2018 – Jan 2024*

Skills: Spring Boot, Selenium, Snowflake, SQL Server, Oracle, ETL Validation

ESP: reporting platform consolidating data from multiple sources into interactive extract reports for clients.

- Enhanced the Spring Boot automation framework to support ongoing module updates for the PC project.
- Built automation to interact with multiple databases (Snowflake, SQL Server, Oracle, ESP) for end-to-end ETL validation, building source-vs-target comparison and summary reporting, cutting manual reconciliation by 55%.
- Developed Selenium automation for the ESP application and its reporting workflows.
- Executed regression suites across Meta Model, Oracle Meta Model, Data Mart, and Extract modules each release.

Programmer Analyst, QA Analyst — Cognizant *May 2015 – Jul 2018*

Skills: TestNG, Selenium WebDriver, Java, SQL

ICON: insurance platform for Auto, Workers' Compensation, and Spectrum policies, from quoting through policy issuance.

- Designed and implemented a TestNG + WebDriver automation framework; performed database and web service testing.
- Authored test scenarios, test cases, and RTM/Compatibility Matrix documentation from business requirements.
- Executed Smoke, Functional, System Integration, Regression, and Retesting cycles; led internal test case peer reviews.

EDUCATION

- Master of Computer Applications (MCA) — University College of Engineering (Osmania Campus), Hyderabad

LANGUAGES

English, Hindi, Telugu